Satellite Data Integration and Processing Framework for Neural Network-Based Precipitation Estimation

Research Team

***Abstract*—Accurate precipitation prediction is crucial for extreme weather mitigation and meteorological modeling. This paper presents a systematic data acquisition, preprocessing, and validation pipeline that integrates high-resolution satellite imagery and radar observations. We leverage the Geostationary Operational Environmental Satellite (GOES) system and the Multi-Radar/Multi-Sensor (MRMS) system to extract key physical parameters—specifically brightness temperature, radiance, and lightning flashes. The processed outputs are validated against official reference datasets, establishing a reliable, high-fidelity data foundation designed for neural network-based precipitation estimation models.**

***Index Terms*—GOES-R, MRMS, Remote Sensing, Data Pipeline, Precipitation Estimation, Neural Networks.**

I. Introduction

Severe weather events and extreme precipitation pose significant challenges to infrastructure, agriculture, and human safety. Advanced satellite and radar systems have significantly improved our capacity to monitor these phenomena in real time. The Geostationary Operational Environmental Satellite (GOES) R-Series, managed by the National Oceanic and Atmospheric Administration (NOAA), provides continuous, high-resolution hemispheric observations. Concurrently, the Multi-Radar/Multi-Sensor (MRMS) system, developed by NOAA's National Severe Storms Laboratory (NSSL), integrates networks of ground-based radars to deliver precise, three-dimensional precipitation estimates.

The primary objective of this project is to construct an end-to-end data pipeline in Python that ingests multi-source meteorological datasets, performs spatial alignment and cropping to target mesoscale areas, and prepares the inputs for assimilation into deep learning architectures. By correlating satellite-derived cloud properties, thermodynamic profiles, and lightning activity with ground-truth radar precipitation rates, the proposed framework establishes a robust dataset tailored for neural network training and validation.

II. Methodology and Data Acquisition

A. Data Sources

Our meteorological pipeline ingests three core physical datasets:

- **Advanced Baseline Imager (ABI):** Captures multispectral imagery of Earth's weather, oceans, and land surface.
- **Geostationary Lightning Mapper (GLM):** A single-channel, near-infrared optical transient detector that continuously measures total lightning activity (both in-cloud and cloud-to-ground) across the Western Hemisphere.
- **Multi-Radar/Multi-Sensor (MRMS):** Merges data from multiple radars, satellites, and gauge networks to generate high-resolution Quantitative Precipitation Estimations (QPE). Specifically, this study utilizes MRMS Pass 2 data for high-accuracy precipitation rates.

This work was supported by the respective meteorological data repositories.

B. Data Retrieval and Synchronization

To handle the massive volume of raw geostationary and radar data, files were retrieved and synchronized to local storage using `rclone`, an open-source command-line tool capable of managing file transfers across diverse cloud storage backends (e.g., Amazon S3, Google Drive). The cross-platform compatibility of `rclone` ensures that the ingestion script remains reproducible and executable across Windows, macOS, and Linux platforms.

C. ABI Spectral Channel Selection

We select five specific infrared channels from the 16 available spectral bands on the ABI instrument to capture atmospheric dynamics and cloud microphysics:

- **Channel 8 ($6.2\,\mu$m):** Upper-level water vapor; used to map high-altitude wind vectors and moisture patterns.
- **Channel 10 ($7.3\,\mu$m):** Lower-level water vapor; tracks mid-to-low-level atmospheric moisture.
- **Channel 11 ($8.4\,\mu$m):** Cloud-top phase; distinguishes between liquid water, supercooled water, and ice crystals.
- **Channel 14 ($11.2\,\mu$m):** Standard longwave infrared window; measures cloud-top and surface skin temperatures.
- **Channel 15 ($12.3\,\mu$m):** Dirty longwave infrared window; assists in split-window moisture corrections and ash/dust detection.

III. Results and Validation

A. Satellite Projection and Regridding

Using custom Python scripts, the raw GOES ABI full-disk data was parsed to extract radiance and compute the Brightness Temperature Index (BTi). To verify the geometric fidelity of the spatial grid, our initial checks validated the complete coverage boundaries of the full-disk dataset.

As shown in Fig. 1, our initial processing confirms the complete geographic bounds under the satellite orthographic projection. Following validation of the outer limits, the data was cropped to a coordinate-aligned subgrid (mesoscale projection) focusing precisely on our target research domain.

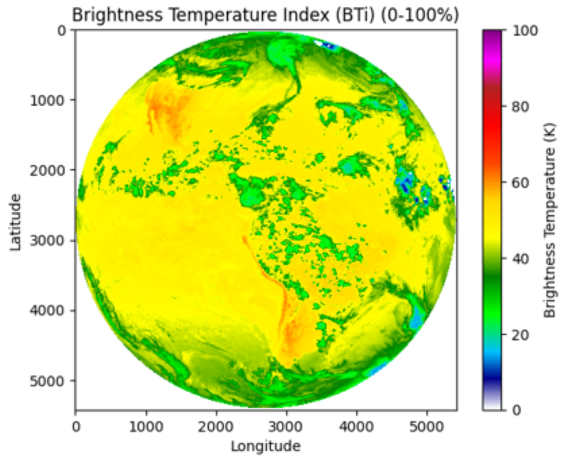

Fig. 1. Verification of spatial coverage and boundary alignment for the raw satellite projection.

B. Lightning Flash Mapping

Using the Geostationary Lightning Mapper (GLM) data, individual lightning occurrences were extracted as spatial coordinates. These coordinates were projected onto our spatial maps to align lightning flashes directly with structural cloud features.

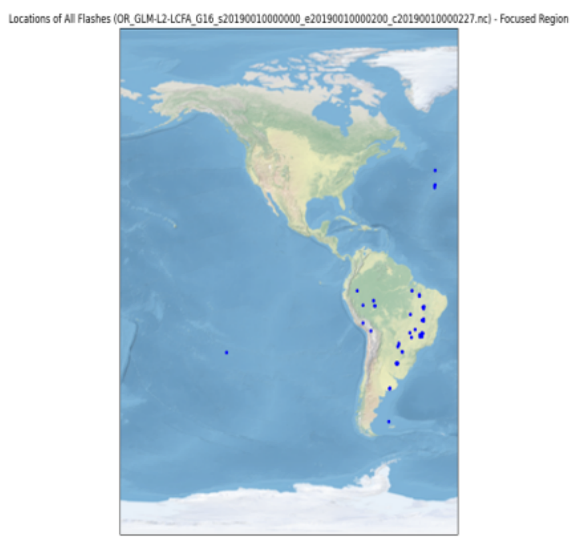

Fig. 2. Mapping of lightning occurrences and storm structures over the wider Caribbean domain.

Fig. 2 displays the compiled lightning flash occurrences over the wide observation range. A more localized view of this distribution is presented in Fig. 3, focusing on convective cloud cells. Rapid spikes in these localized lightning densities act as precursor elements to intense localized updrafts and severe rainfall, providing vital spatial input features for our deep learning networks.

C. Radar Precipitation Mapping and Target Selection

Puerto Rico was chosen as our primary localized study region due to its vulnerability to extreme tropical precipitation systems and its consistent multi-sensor radar coverage. Using MRMS data, we isolated the precipitation rate over the island.

Fig. 3. High-resolution spatial distribution of GLM lightning flash coordinates across the localized observation window.

The data was cropped and visualized under a regional mesoscale projection defined by Latitude $18.0°$N to $18.5°$N and Longitude $-67.2°$W to $-65.8°$W.

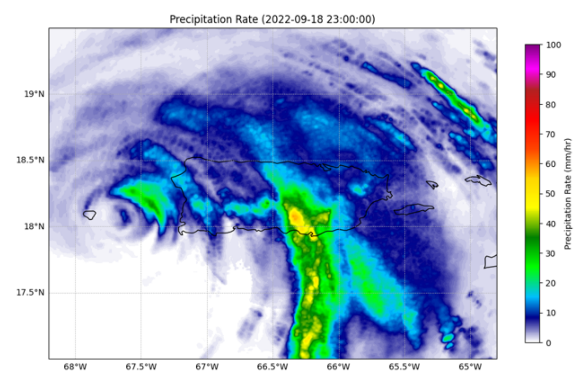

Fig. 4. Visualized Brightness Temperature Index (BTi) and mesoscale precipitation rate profiles.

Fig. 4 highlights the calculated Brightness Temperature Index (BTi) alongside corresponding spatial precipitation rates across our target mesoscale domain. Additionally, a three-dimensional topographic projection was generated using the precipitation rate along the vertical z-axis to analyze the spatial structure of localized convective cells.

D. Ground-Truth Cross-Validation

To verify the fidelity of the pipeline, our processed datasets were evaluated against NOAA's official MRMS Product Viewer. Direct pixel-to-pixel comparisons of precipitation rates for historical storm events (e.g., September 18, 2022) demonstrated exact alignment. This successful cross-validation confirms that our Python scripts accurately handle georeferencing, coordinate transforms, and unit scaling.

IV. Conclusion

This study successfully implemented and validated a multi-sensor data preprocessing pipeline tailored for machine learning models. By integrating geostationary satellite imagery (GOES ABI), near-real-time lightning observations (GLM), and high-resolution radar precipitation rates (MRMS), we generated structured, spatially aligned training datasets. The close alignment between our processed outputs and official NOAA products establishes a mathematically sound foundation. Future work will ingest these aligned spatial matrices into convolutional neural networks (CNNs) and recurrent architectures (e.g., ConvLSTM) to perform short-term, high-resolution precipitation forecasting.

References

[1] NOAA/NASA, "GOES-R Series Space Segment," *GOES-R Series Program Office*, https://www.goesr.gov.

[2] National Severe Storms Laboratory, "Multi-Radar/Multi-Sensor (MRMS) System," *NOAA NSSL*, https://mrms.nssl.noaa.gov.